Adaptivity of Diffusion Models to Manifold Structures

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Abstract

Empirical studies have demonstrated the effectiveness of (score-based) diffusion models in generating high-dimensional data, such as texts and images, which typically exhibit a low-dimensional manifold nature. These empirical successes raise the theoretical question of whether score-based diffusion models can optimally adapt to low-dimensional manifold structures. While recent work has validated the minimax optimality of diffusion models when the target distribution admits a smooth density with respect to the Lebesgue measure of the ambient data space, these findings do not fully account for the ability of diffusion models in avoiding the the curse of dimensionality when estimating high-dimensional distributions. This work considers two common classes of diffusion models: Langevin diffusion and forward-backward diffusion. We show that both models can adapt to the intrinsic manifold structure by showing that the convergence rate of the inducing distribution estimator depends only on the intrinsic dimension of the data. Moreover, our considered estimator does not require knowing or explicitly estimating the manifold. We also demonstrate that the forward-backward diffusion can achieve the minimax optimal rate under the Wasserstein metric when the target distribution possesses a smooth density with respect to the volume measure of the low-dimensional manifold.

1 Introduction

Generative models have emerged as powerful and routinely utilized tools for generating complex data, finding numerous applications across various domains, including computer vision Park et al. (2021); Wang et al. (2021); Turhan and Bilge (2018), natural language processing Salakhutdinov (2015); Nadkarni et al. (2011), and bioinformatics Cheng et al. (2021); Lan et al. (2020). Contrasted with classical explicit distribution estimation approaches, generative modeling implicitly estimates the data distribution by characterizing the data-generating process, and can adeptly capture highly nonlinear structures that may lead to singularities, such as jumps and point masses, in the distribution. Additionally, generating samples from the underlying data distribution can be more useful and important than estimating it in many applications, such as synthetic image creation, automated text generation, and biological structure simulation.

Various architectures and training methodologies, such as Generative Adversarial Networks (GAN) Goodfellow et al. (2014), Variational Autoencoders (VAE) Kingma and Welling (2013), and flow-based generative model Papamakarios et al. (2021), have been developed to enhance the efficacy and application range of generative models, each presenting unique strengths and challenges. Recently, a new class of generative models, known as (score-based) diffusion models Ho et al. (2020); Song et al. (2020); Nichol and Dhariwal (2021); Song and Ermon (2019), has showcased state-of-the-art performance in various domains, including high-quality image generation Song et al. (2020); Nichol and Dhariwal (2021), photorealistic text-to-image translation Saharia et al. (2022), and high-fidelity audio production Kong et al. (2020). In particular, two classes of diffusion models are prevalently employed for sampling and data generation. One is Langevin diffusion models, which leverage Langevin dynamics to gradually transition a simple initial distribution to the target data distribution, making use of the gradient of the logarithmic data density (i.e. score), typically estimated through score matching. The other is forward-backward diffusion models, which employ two diffusions to construct the generative model. The first diffusion, called the forward process, utilizes an analytically tractable stochastic differential equation, such as the Ornstein-Uhlenbeck (OU) process, to trans-

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form the data distribution to a simple noise distribution. The second diffusion, called the backward process, utilizes the time-reversal of the forward process to generate data from noise based on the score estimated from the forward process.

Despite the high-dimensional form of data in various applications, the empirical success of state-of-the-art generative modeling approaches is often attributed to the identification and utilization of low-dimensional manifold structures within the data. These structures enable a means to circumvent the curse of dimensionality, allowing generative models to adeptly adapt to manifold structures. For instance, earlier generative modeling approaches, including GAN and VAE, typically involve the extraction of latent features or representations (encoding) that are used for accurately reconstructing the original data (decoding). In other words, a low-dimensional manifold structure is implicitly assumed and utilized in distribution modeling and estimation. In contrast, diffusion models do not explicitly estimate or utilize the manifold structure, beyond merely injecting Gaussian noise to smooth out the (possibly) singular data distribution, yet they achieve remarkably accurate data generation. Motivated by these considerations, the present study aims to address the following theoretical question: *Is di usion modeling able to optimally adapt to the manifold structure in the data? In other words, does the convergence rate of the induced distribution estimator from di usion models depend only on the intrinsic dimension of the data, and is the rate optimal?*

Related works. Recently, convergence rates of generative models for implicit distribution estimation have been investigated by a number of works. Tang and Yang (2021) examines the excess risk associated with VAE through the lens of M-estimation. When specialized to Gaussian encoders and decoders with mean functions approximated by ReLU neural networks, their result demonstrates that VAE can adapt to low-dimensional manifold structures. However, the derived rate of convergence is worse than the minimax-optimal rate, as the KullbackLeibler (KL) divergence objective in VAE appears unsuitable for comparing mutually singular distributions. Several recent studies establish quantitative convergence rates for GAN in distribution estimation under various discrepancy metrics, such as the Jensen-Shannon divergence Belomestny et al. (2021), Wasserstein distances Liang (2021); Chae (2022); Tang and Yang (2023), and adversarial losses (also termed integral probability metrics) Liang (2021); Tang and Yang (2023); Uppal et al. (2019). Among these, Liang (2021) demonstrates that, by replacing the empirical distribution with a regularized version that incorporates the smoothness of the target density function, GAN can attain the minimax rate of convergence for smooth density estimation under the 1-Wasserstein metric. Furthermore, Tang and Yang (2023) establishes the minimax rate under adversarial losses for estimating smooth distributions supported on manifolds, and shows that a regularized GAN explicitly incorporating the manifold structure can attain this rate.

For Langevin diffusion models, some previous works such as Huggins and Zou (2017); Dalalyan and Karagulyan (2019); Yang and Wibisono (2022) have studied its convergence and asymptotic bias due to the use of an inaccurate score (e.g., based on stochastic gradient or score matching). However, their assumptions on the score approximation either requires a nearly L_1-accurate score estimator Dalalyan and Karagulyan (2019), or a controlled moment generating function for the approximation error Yang and Wibisono (2022), both implying a controlled error under all finite moments. In comparison, our proof only requires a fourth-moment error bound (expectation under the stationary distribution of the diffusion) on the score estimation.

For forward-backward diffusion models, Chen et al. (2022); Lee et al. (2023) demonstrate that an L_2-accurate score estimator leads to a controlled distribution estimation error bound in the total variation distance. Oko et al. (2023) analyzes the L_2-error in score estimation utilizing score matching over a neural network class, and demonstrates that, under certain smoothness conditions on the true density function, the estimated data distribution achieves the minimax optimal rate both in the total variation distance and in the 1-Wasserstein distance. For data distributions supported on manifolds, Pidstrigach (2022) identifies conditions that enable forward-backward diffusion to generate samples from the data manifold and highlights the drift explosion in the backward diffusion process as time progresses; De Bortoli (2022) examines convergence in the 1-Wasserstein distance under an L_2 error assumption on the score estimator; and Oko et al. (2023); Chen et al. (2023) establish explicit convergence rates using specific score estimation methods when the data-supporting manifold is a low-dimensional hyperplane in the ambient space, with the rate by Oko et al. (2023) attaining the minimax optimality in the 1-Wasserstein distance.

Our contributions. In this paper, we illustrate that both diffusion models can adapt to the intrinsic manifold structure by demonstrating that the convergence rates of the inducing distribution estimators are $n^{-O(d^{-1})}$ up to logarithmic terms, with d denoting the data intrinsic dimension. Interestingly, unlike other generative modelling approaches such as GAN and VAE, our considered estimator does not need

knowing or explicitly estimating the manifold. Furthermore, our result shows that the forward-backward diffusion can achieve the minimax optimal rate of $\max\{n^{-\frac{1}{d+4}}, n^{-\frac{\beta+1}{2\beta+d}}\}$ under the 1-Wasserstein metric when the target distribution admits an β-smooth density with respect to the volume measure of a (potentially non-linear) d-dimensional manifold in the ambient space R^D. For Langevin diffusion models, in order to appropriately define the drift based on a singular data distribution, we consider a Gaussian-smoothed score and a corresponding score estimation method; technically, we demonstrate that a fourth-moment error bound on the score estimator suffices to imply a distribution estimation error bound, which refines existing theory that assumes either an L_∞ error bound or a moment-generating function bound on the error distribution of the score estimator. For forward-backward diffusion models, we show that the minimax optimal estimation error can be attained without explicitly estimating the manifold by employing a new class of score approximating neural network class whose complexity gradually changes with time t, and derive an explicit score approximation error bound.

2 Diffusion Models and Score Estimation

In this section, we review two representative score-based diffusion models for distribution estimation. We also discuss their adaptations for handling singular distributions with manifold structure.

2.1 Langevin diffusion models

In generative modeling, the goal is to implicitly learn the underlying data distribution p_{data} on data space $X \subseteq \mathrm{R}^D$ by specifying a data generative model that produces samples looking similar to a given set of i.i.d. samples $\{x_i\}_{i=1}^n$ from p_{data}. Earlier attempts (e.g., Song and Ermon (2019)) to address this problem using diffusion models directly used a (time-discretized) Langevin model to generate new data when p_{data} admits a density with respect to the Lebesgue measure on R^D,

$$\mathrm{d}X_t = \nabla \log p_{\mathrm{data}}(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \quad X_0 \sim p_0, \quad (1)$$

where $\{B_t : t \geq 0\}$ denotes the standard Brownian motion in R^D, p_0 is an initial distribution that is easy to sample from, and $\nabla \log p_{\mathrm{data}} : \mathrm{R}^D \to \mathrm{R}^D$ is called the score function defining the drift term of the diffusion model. As a well-known result, the stationary (or limiting) distribution of the Langevin model (1) coincides with the target distribution p_{data}. In other words, the distribution p_t of X_t converges to p_{data} as $t \to \infty$ under various metrics over $\mathscr{P}(X)$, the space of all distribution on the data space $X \subseteq \mathrm{R}^D$. In practice, the score function needs to be estimated; we defer details about score estimation using the finite sample set $\{x_i\}_{i=1}^n$ to Section 2.3. In this paper, we aim to keep the presentation simple by ignoring the technical issues that arise from the time-discretization error in simulating or generating samples from diffusion models, which have been addressed in many existing works, e.g., Zhang et al. (2023); Dalalyan (2017); Li et al. (2019). Unfortunately, this conceptually simple score-based diffusion modeling approach has a notable drawback: the convergence of p_t to its limit p_{data} can be exponentially slow due to the non-log-concavity or multi-modality of p_{data}.

When dealing with high-dimensional data residing on low-dimensional manifolds, a common scenario in image and text generation, p_{data} becomes a singular distribution on the data ambient space R^D. In such cases, Song and Ermon (2019) proposes an annealing approach, where they use scores associated with the Gaussian-smoothed data distribution $p_{\mathrm{data},\sigma}(\cdot) = \int_{\mathrm{R}^D} p_{\mathrm{data}}(y)\,\phi_\sigma(\cdot - y)\mathrm{d}y$ with different levels of noise to construct a sequence of annealed Langevin models. Here, ϕ_σ denotes the density function of $N(0, \sigma^2 I_D)$. In the sampling stage, noise levels are gradually decreased as the sampling process approaches the data manifold. In this work, we instead consider the following Gaussian-smoothed Langevin diffusion

$$\mathrm{d}X_t = \nabla \log p_{\mathrm{data},\sigma}(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}B_t, \quad X_0 \sim p_0 \quad (2)$$

using a single noise parameter σ to optimally trade-off the bias and variance in order to attain a best estimation error. Intuitively, this parameter σ plays a similar role as an inverse bandwidth parameter as in the kernel density estimator (e.g., Kim et al. (2019); Divol (2022) for KDE on manifolds). The first contribution of this paper is to show that, with a properly chosen σ that depends only on the sample size n and the intrinsic dimensionality d of the data, this Gaussian-smoothed Langevin diffusion can adapt to the intrinsic manifold structure by showing that the convergence rate of the inducing distribution estimator for estimating p_{data} depends only on d. Here, the estimation of the noise-perturbed score function $\nabla \log p_{\mathrm{data},\sigma}$ is discussed in Section 2.3.

2.2 Forward and backward diffusion models

To address the issue of potentially exponentially slow convergence inherent to the Langevin diffusion model, several recent papers (e.g., Ho et al. (2020); Song et al. (2020)) have introduced forward and backward diffusion models. These strategies employ two diffusion processes collaboratively: one for constructing more com-

plex, time-dependent score functions, and the other for generating samples through a time-inhomogeneous process, based on the estimated score functions. Consequently, this method can circumvent the slow convergence typically associated with using a single diffusion model.

Specifically, the first diffusion process, referred to as the forward diffusion, employs a simple diffusion starting from p_{data} that admits a closed-form solution and converges exponentially quickly to its limiting distribution, such as the OrnsteinUhlenbeck (OU) process:

$$\mathrm{d}\overset{\prime}{X}_t = -\beta_t \overset{\prime}{X}_t \mathrm{d}t + \sqrt{2\beta_t}\mathrm{d}B_t; \quad \overset{\prime}{X}_0 \sim p_{\text{data}}; \tag{3}$$

for some (possibly time-dependent) drift coefficient $\beta_t : t \geq 0$ and scalar diffusion coefficient $\sqrt{2\beta_t} : t \geq 0$. Without loss of generality, we will focus on the OU process with $\beta_t = \sqrt{2\beta_t}$ as the forward diffusion in this paper,[1] which admits the closed form solution $X_t = m_t X_0 + \int_0^t \frac{m_t}{m_s}\sqrt{2\beta_s}\,\mathrm{d}B_s$ and has the conditional distribution of $p_t(\cdot \mid X_0) = N(m_t X_0, \sigma_t^2 I_D)$ given X_0, where $m_t = \exp\left(-\int_0^t \beta_s \mathrm{d}s\right)$ and $\sigma_t^2 = 1 - m_t^2$. For example, for constant drift $\beta_t \equiv \beta$ and diffusion $\sqrt{2\beta}$, we have $m_t = \exp(-\beta t)$, $\sigma_t^2 = 1 - \exp(-2\beta t)$, and p_t converges exponentially quickly to its limiting distribution $p_\infty = N(0, \sigma_\infty^2 I_D)$ with $\sigma_\infty^2 = 1$ under the total variation metric d_{TV}, or

$$d_{\text{TV}}(p_t, p_\infty) \leq C\exp(-\beta t), \quad t \geq 0; \tag{4}$$

for some constant C only depending on $p_0 = p_{\text{data}}$. Using sample trajectories generated from the forward diffusion (3), one can estimate the (time-dependent) score function $\nabla \log p_t : \mathbb{R}^D \to \mathbb{R}^D$ by score matching (c.f. Section 2.3), where p_t denotes the (unconditional) distribution of X_t, for t from zero to a sufficiently large time $T \asymp \log(\varepsilon^{-1})$ such that $d_{\text{TV}}(p_T, p_\infty) \leq \varepsilon$ for some error tolerance level $\varepsilon \in (0, 1)$.

The second diffusion process, usually called the backward diffusion, reverses the forward diffusion:

$$\mathrm{d}X_t = \beta_{T-t} X_t + 2\beta_{T-t}\nabla \log p_{T-t}(X_t)\,\mathrm{d}t + \sqrt{2\beta_{T-t}}\mathrm{d}B_t; \quad X_0 \sim p_T: \tag{5}$$

Under mild conditions on p_{data} Song et al. (2020); Haussmann and Pardoux (1986) (valid for our setting), the distribution of X_t is p_{T-t}, so that $X_T \sim p_0 = p_{\text{data}}$. Since p_T is close to $p_\infty = N(0, I_D)$, one can instead initialize the backward diffusion using the easy-to-sample distribution p_∞, i.e. set $X_0 \sim N(0, I_D)$.

[1] This process is also called Variance Preserving Stochastic Differential Equation (VPSDE) in Song et al. (2020), which yields a process with a fixed variance of one when the initial distribution has unit variance. The analysis of this process is also considered in Oko et al. (2023); Chen et al. (2022).

The drift term of the backward diffusion depends on the score function estimated using the forward diffusion; therefore, the forward and the backward diffusions together yield a generative model for sampling from p_{data}.

When p_{data} is a singular distribution on $\mathbb{R}^D$, the distribution p_t of $\overset{\prime}{X}_t$ for any $t > 0$ from the forward diffusion is the convolution of a rescaled p_{data} and Gaussian noise $N(0, \sigma_t^2 I_D)$, making it absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^D$. Therefore, unlike the Langevin diffusion (1) that requires deliberately injecting Gaussian noise to smooth out p_{data}, the forward and backward diffusion model does not require this extra step. The second contribution of this paper is to show that the forward and backward diffusion model can also achieve the minimax-optimal convergence rate for estimating p_{data}. Moreover, compared to the Langevin diffusion model, the forward and backward diffusion model does not impose any log-concavity condition or any logarithmic Sobolev inequalities on p_{data}. This is consistent with the key observations made in earlier studies (e.g., Chen et al. (2022); Lee et al. (2023)) that do not involve manifold structures.

2.3 Score estimation

Langevin diffusion model: The score function in the Langevin diffusion model can be estimated by score matching Song and Ermon (2019); Vincent (2011). At the population level, score matching solves the following optimization problem

$$\min_{\theta} \mathrm{E}_{x \sim p_{\text{data}}} \| S_\theta(x) - \nabla \log p_{\text{data}}(x) \|^2; \tag{6}$$

where $S_\theta : \mathbb{R}^D \to \mathbb{R}^D$ denotes a score approximating map parameterized by parameter θ, e.g., (deep) neural networks with controlled depth and number of non-zero parameters. Recall that the primary focus of this paper is on estimating a singular distribution with manifold structure. Therefore, we consider using $\hat{S} = S_{\hat{\theta}}$ to approximate the noise-injected score $\nabla \log p_{\text{data},\sigma}$, where $\hat{\theta}$ minimizes the following sample-level score matching loss:

$$\frac{1}{n}\sum_{i=1}^{n} \mathrm{E}_{x \sim p(\cdot \mid x_i)} \| S_\theta(x) - \nabla \log p_\sigma(x \mid x_i) \|^2: \tag{7}$$

Here, $p_\sigma(x \mid x_i) = N(x_i, \sigma^2 I_D)$ denotes the conditional distribution of the Gaussian error-injected random variable x given i-th data x_i, so that the (unconditional) distribution of x is $p_{\text{data},\sigma}$. Finally, the distribution estimator of p_{data} based the estimated Langevin diffusion model is $\hat{p} = \hat{p}_T$, where $\hat{p}_t$ is the distribution